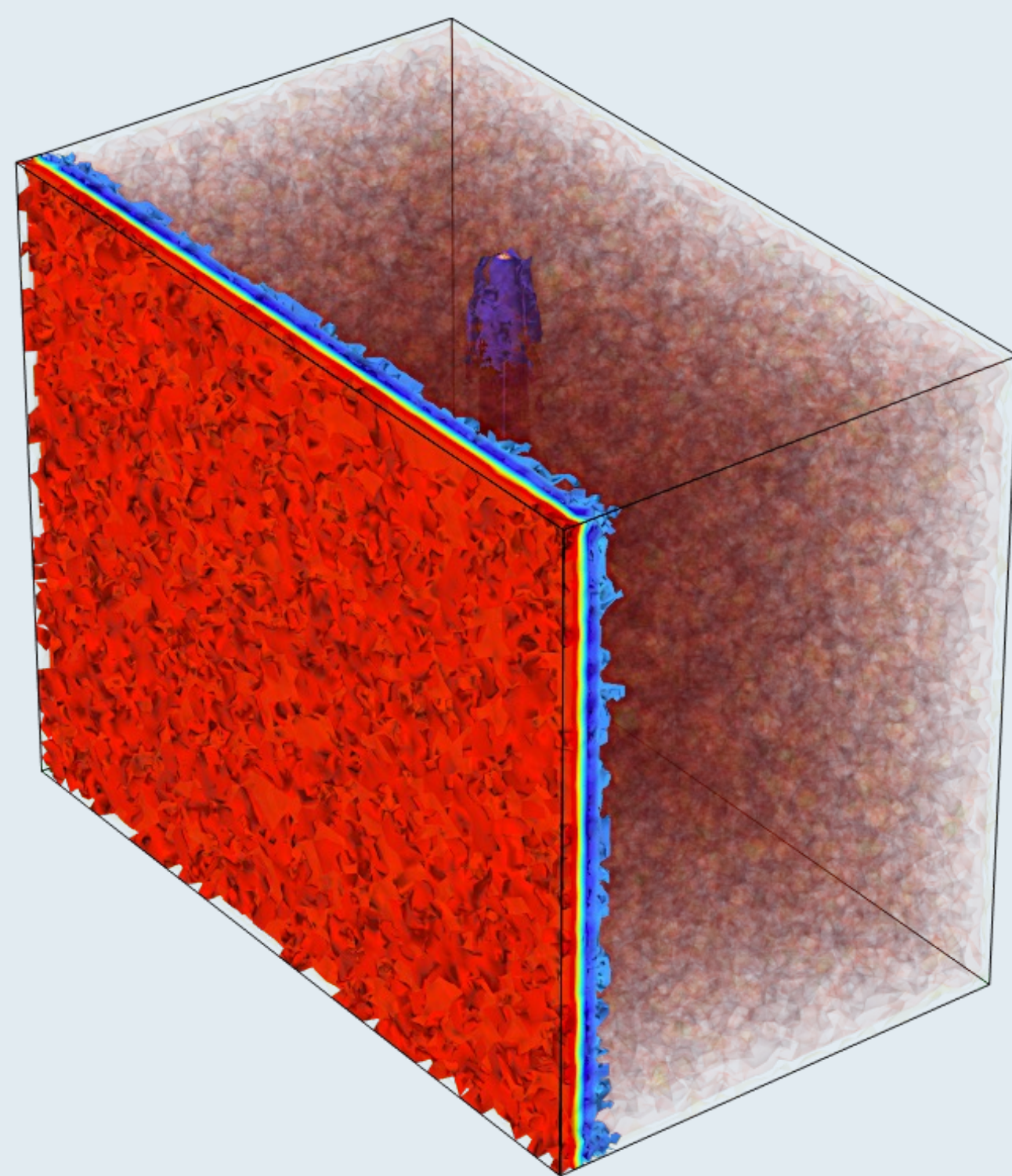




Finite-Element Computational Analysis of Ultrasonic Detection Array Setup for Ultra-High Dose Rate (FLASH) Measurements and Absolute Dosimetry

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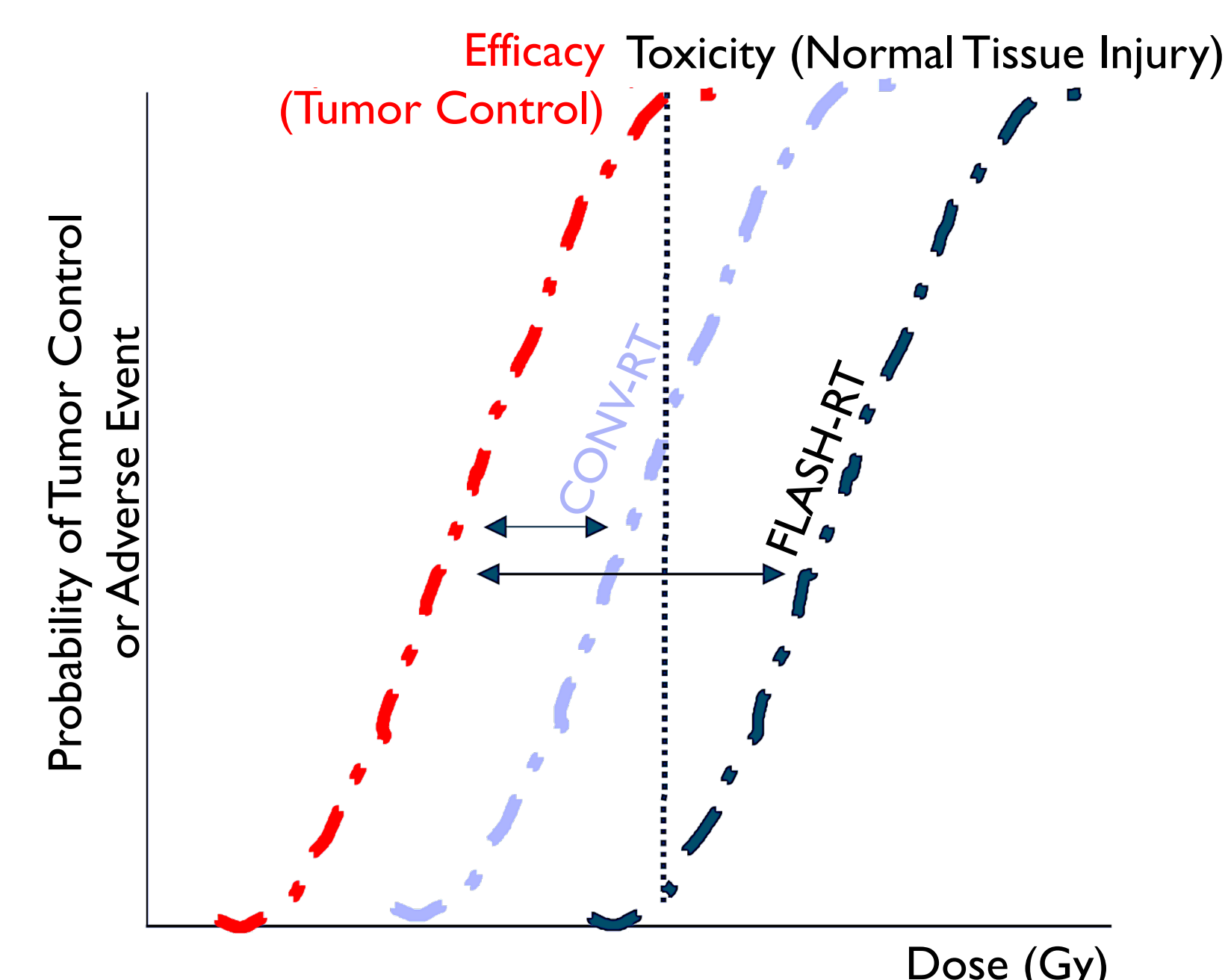
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BACKGROUND



FLASH Radiotherapy (FLASH-RT) delivers radiation dose at an ultra-high dose rate (≥ 40 Gy/s in under 200 ms) in relatively few fractions, compared to Conventional Radiotherapy (CONV-RT), delivering dose at 0.01 – 0.40 Gy/s dose rates for over a minute in a high number of fractions (Tang et al., 2024, p. 2). FLASH-RT minimizes irradiation damage to healthy tissue around a tumor, unlike CONV-RT, by causing metabolic quiescence in normal cells (Limoli & Vozenin, 2023, pp. 6–15).

Fig. 1: Cengel et al., 2024, p. 219.

PURPOSE

Verifying dose rate within a volume for FLASH-RT is critical for clinical adoption, but not possible with current dosimetry hardware or measurement standards. **This work seeks to address both shortcomings by developing an absolute dose and dose-rate detection principle based on ultrasound.** Specifically, by a Finite-Element Method (FEM) analysis of the proposed detection setup via simulations of a complete thermoacoustic model on the COMSOL® Multiphysics 6.1 software.

APPROACH

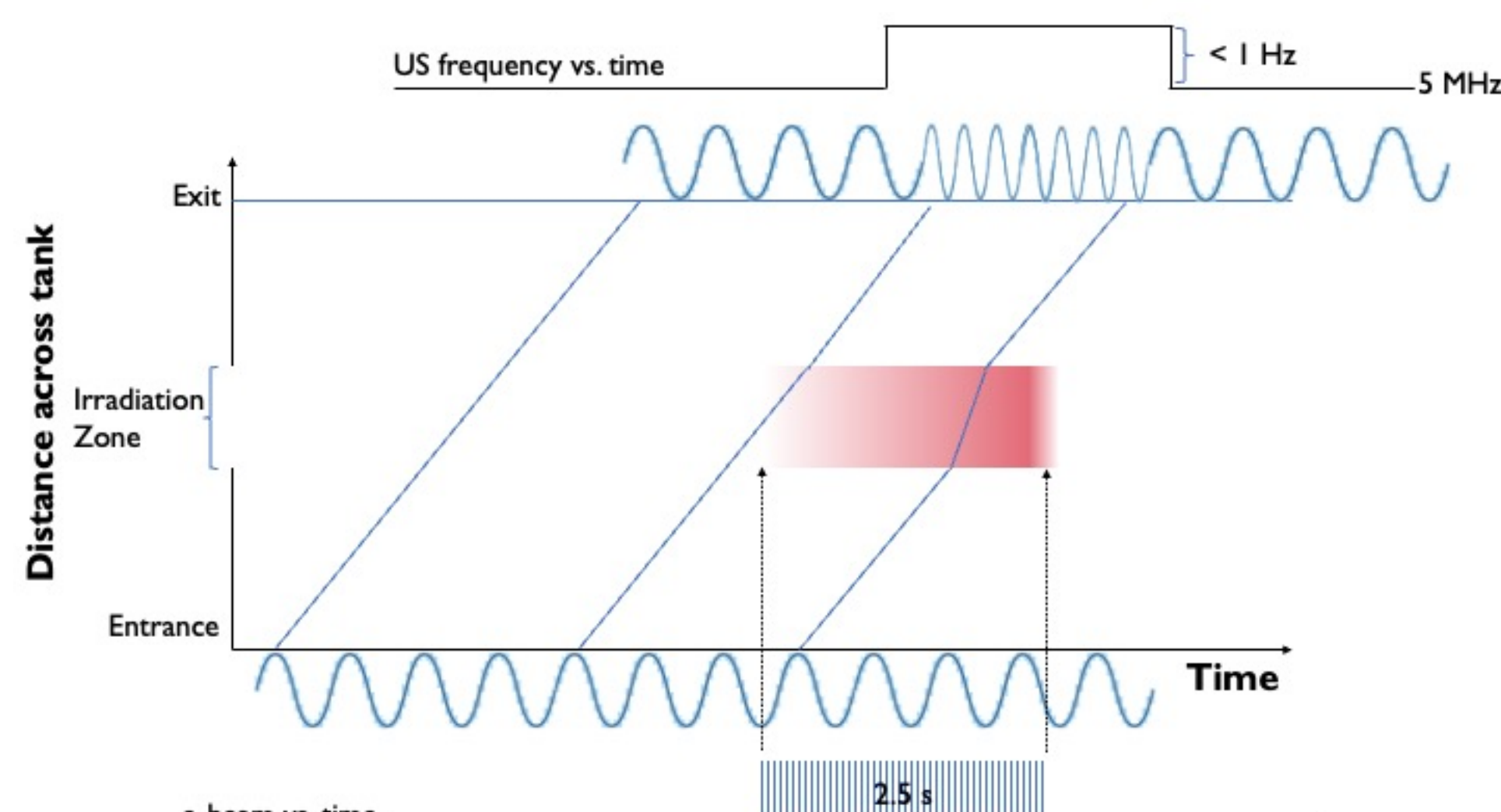
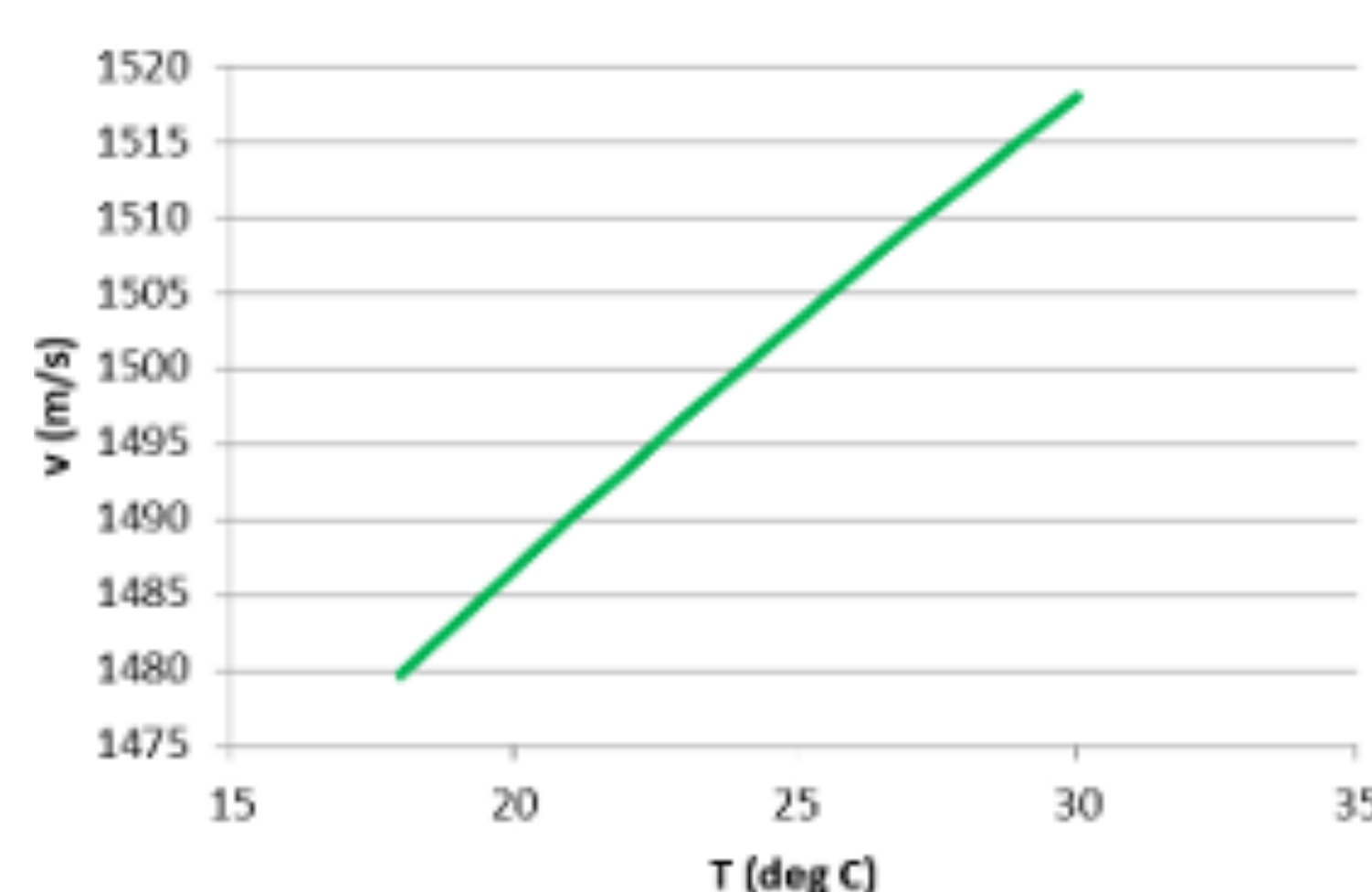
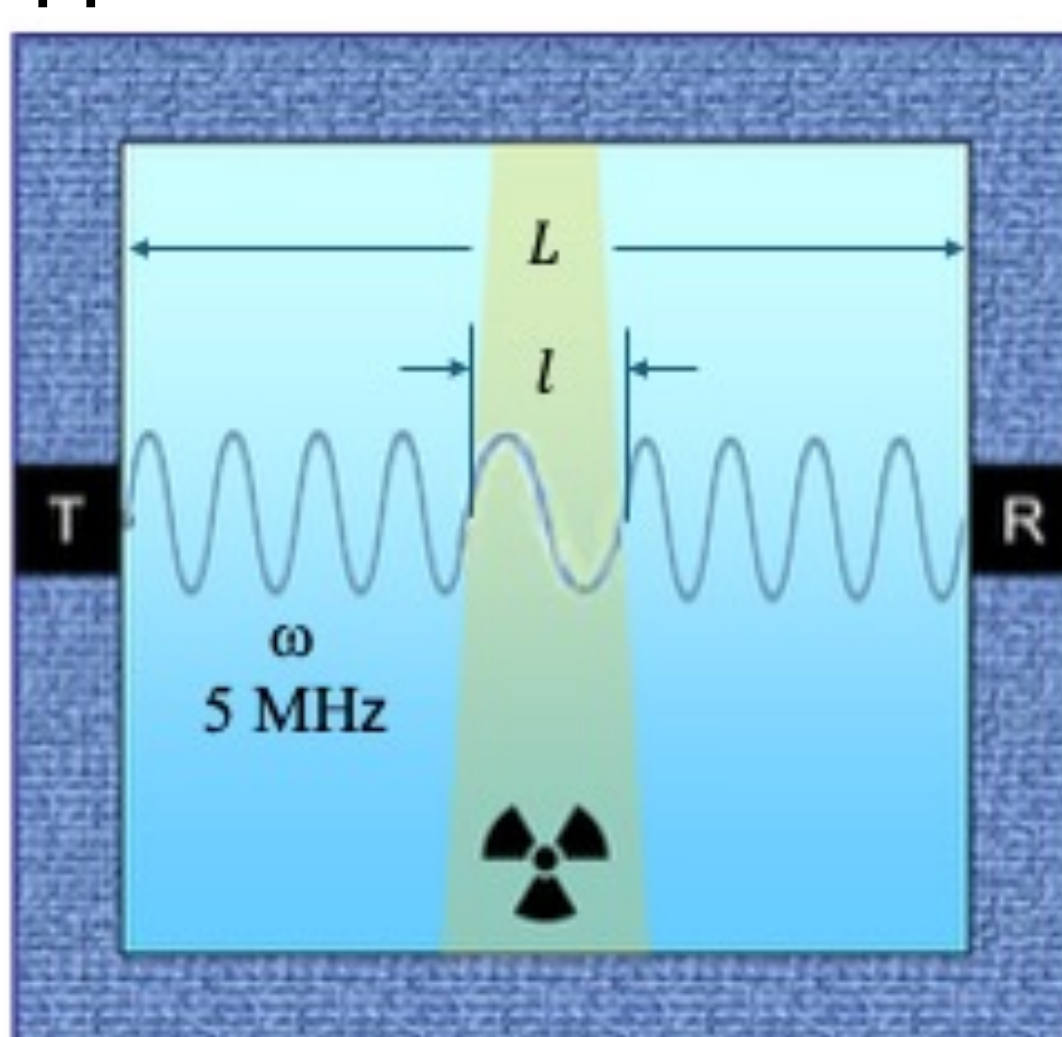


Fig. 2 (Left): Speed of sound in water increases with temperature. Fig. 3 (Right): Radiation heating of water produces a time-dependent phase shift in ultrasonic waves. Periodic application of radiation therefore frequency-modulates the injected carrier of 5 MHz.



PROPOSED SETUP

Fig. 4: Propagate an ultrasonic wave orthogonal to a radiative beam in a water phantom to observe potential phase modulation of the ultrasound for de-modulation, determining the delivered dose rate.

MODEL VALIDATION

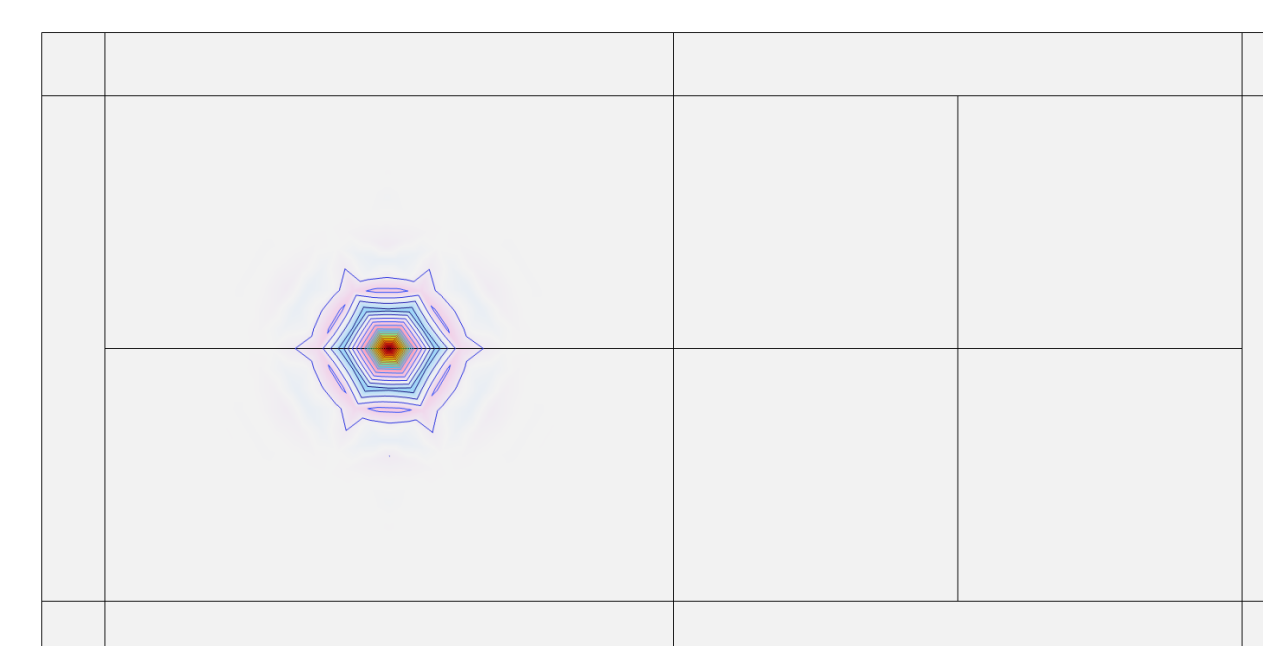


Fig. 5: Simulation of ultrasonic signal, modified from a High-Intensity Focused Ultrasound (HIFU) macro, at 1 μ s. Dimensions modeled after "fish-tank" bath used in experiment.

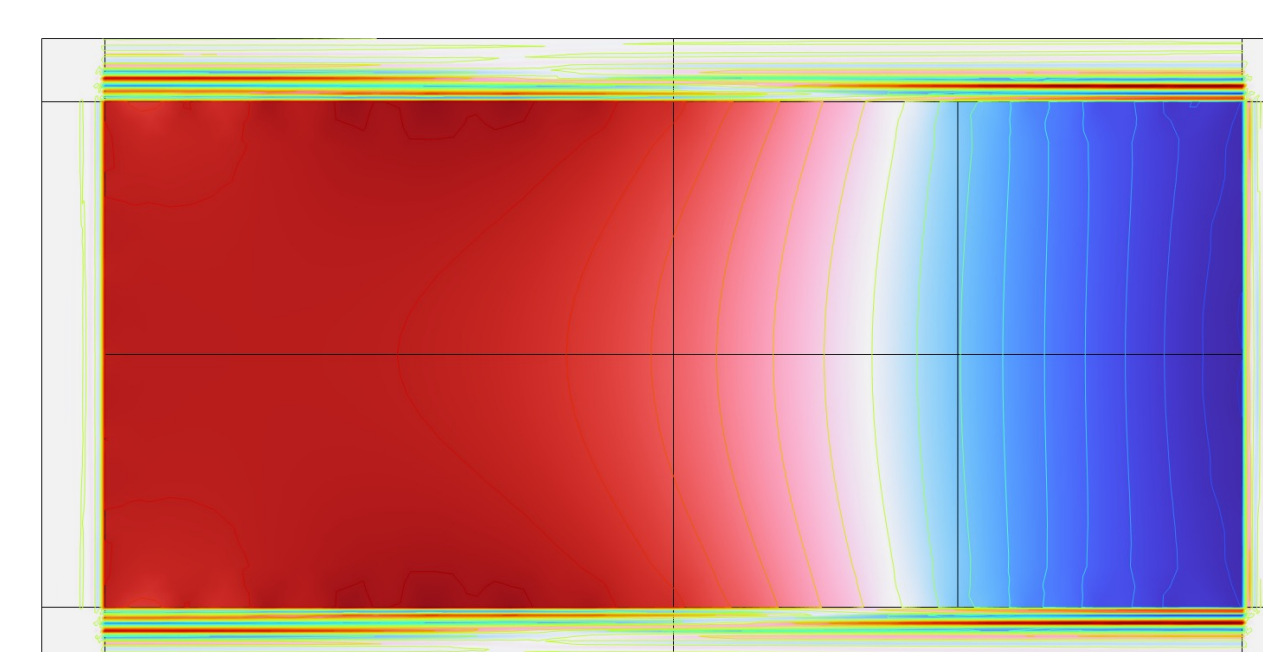


Fig. 6: Simulation of the same ultrasonic signal at 60 μ s. Localization of normalized Acoustic Pressure in the less-dense heated region is shown.

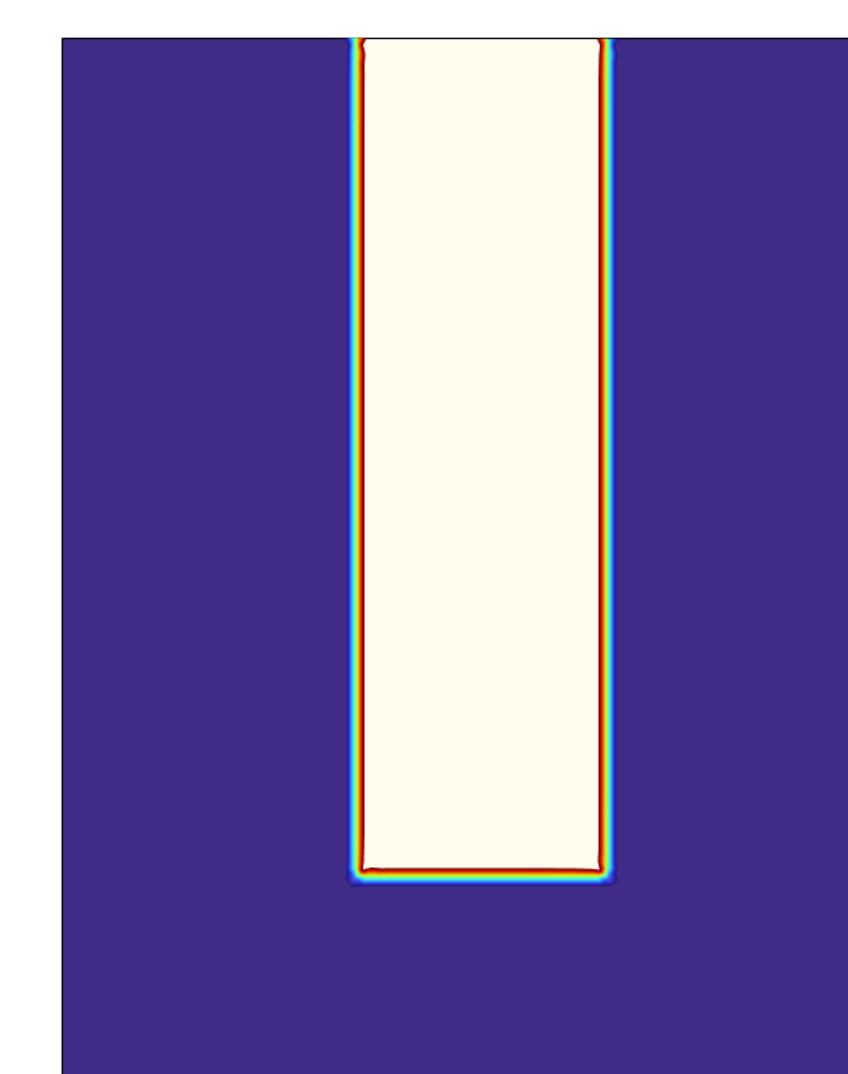
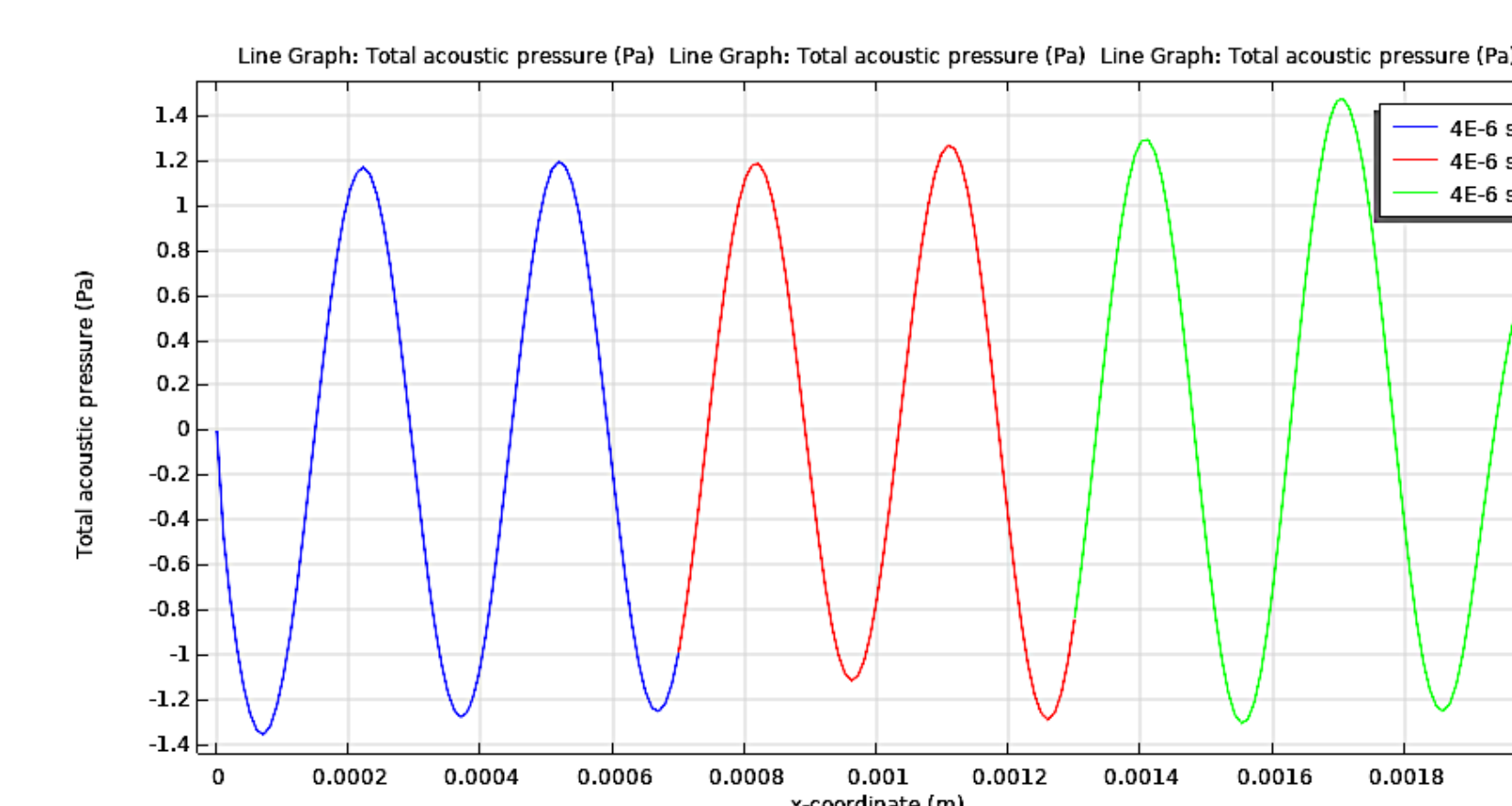
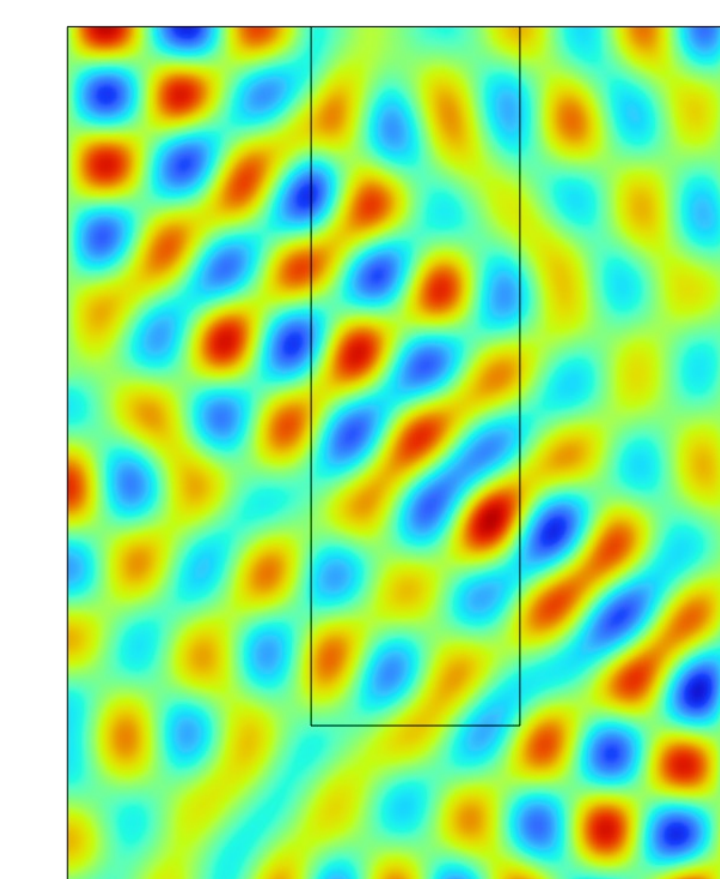
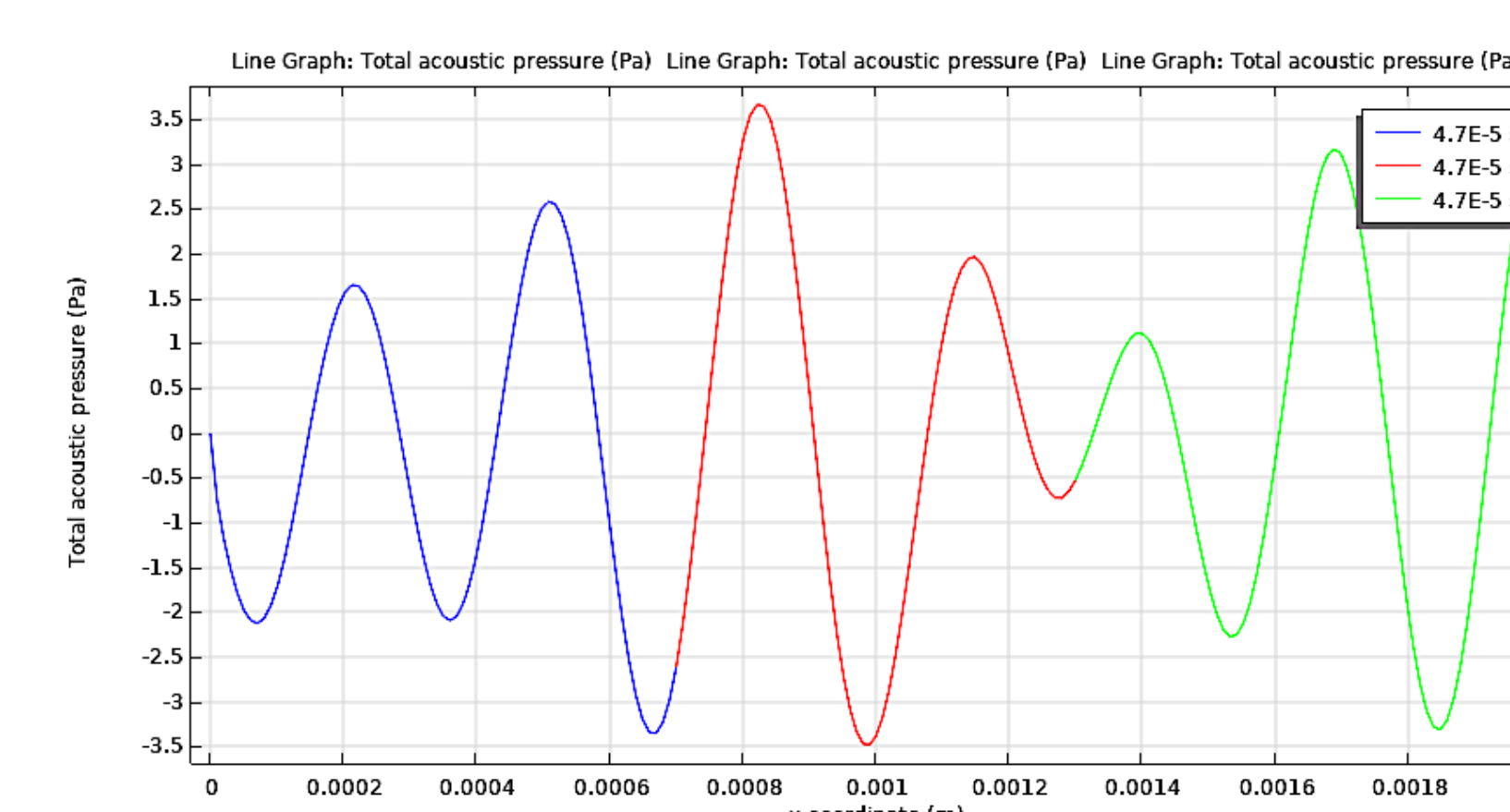
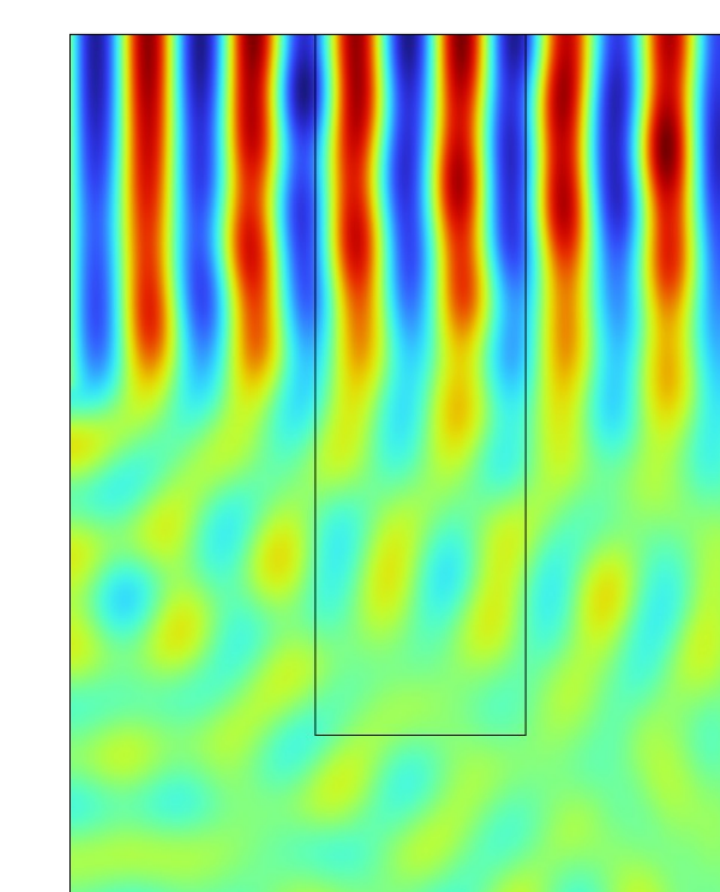
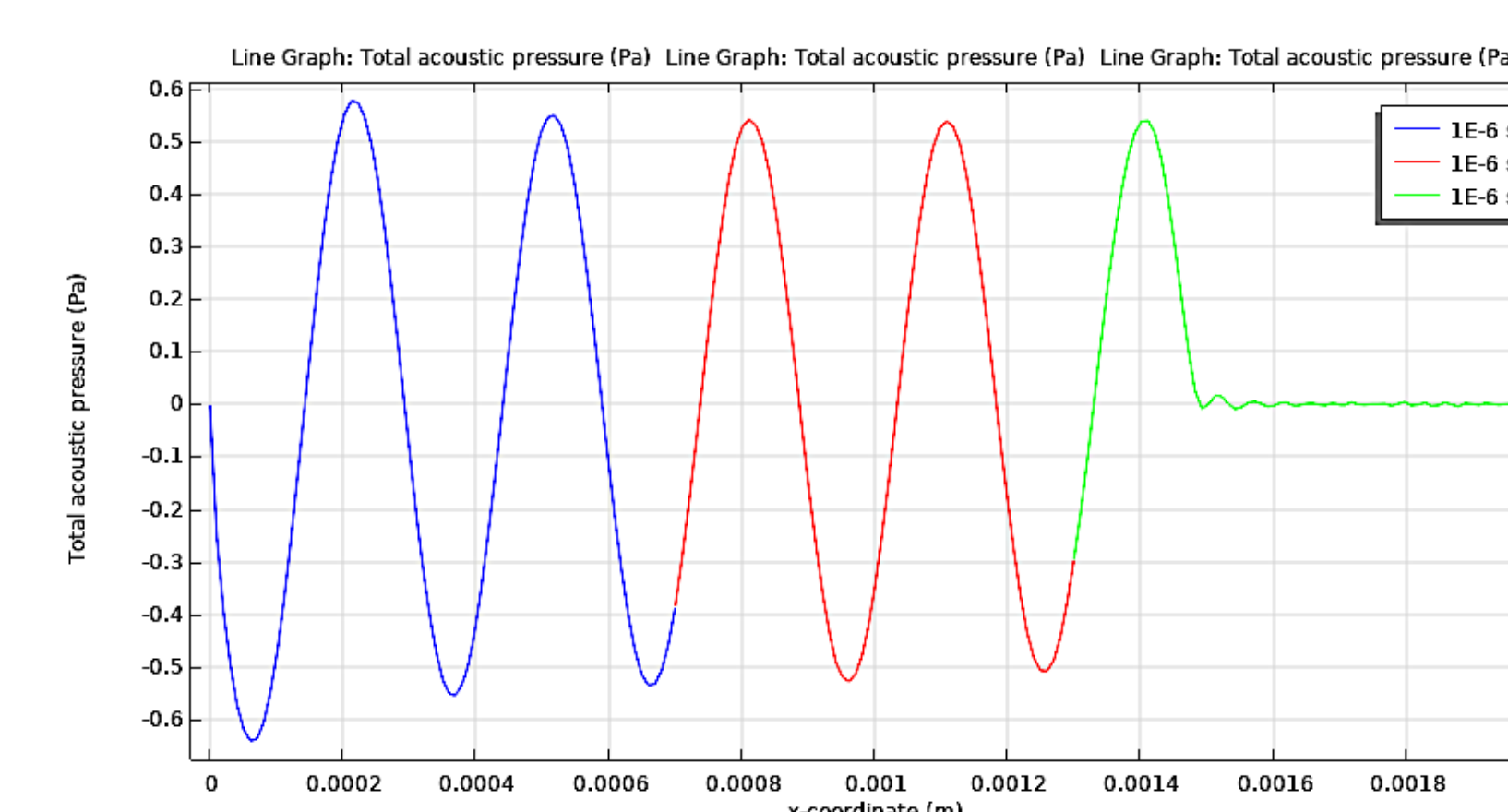
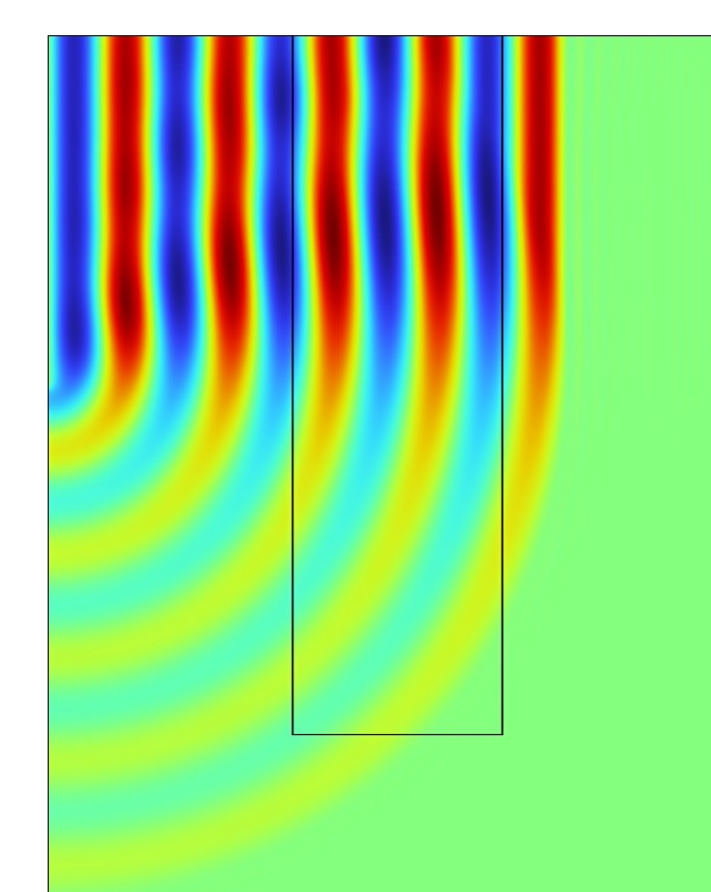


Fig. 7: Temperature surface plot, with a vertical radiative beam modeled by a Heat Flux condition coupled with the Radiative Beam sub-module. Sub-Kelvin heating of a FLASH-RT dose (approximately 10 mK temperature difference achieved by a 40 Gy/s dose rate beam) is shown.

RESULTS



Normalized Acoustic Pressure coupled with Heat Transfer: Fig. 8 (Upper Left): Propagation at 1 μ s. Fig 9 (Upper Right): Propagation at 4 μ s. Fig 10 (Lower Left): Propagation at 47 μ s. Fig 11 (Lower Right): Propagation at 60 μ s.

CONCLUSION

The simulation results show a significant phase modulation of the ultrasonic signal after reaching the radiative beam. Upon interaction with the heated region, the ultrasonic signal initially faced an increase in frequency via compression before decreasing below the original frequency. A decrease in the number of peaks was additionally observed, indicating phase modulation. Moreover, slight amplitude modulation in the heated region was observed in the transient domain. These simulated results are also validated by preliminary results observed in experiment with close agreement.

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